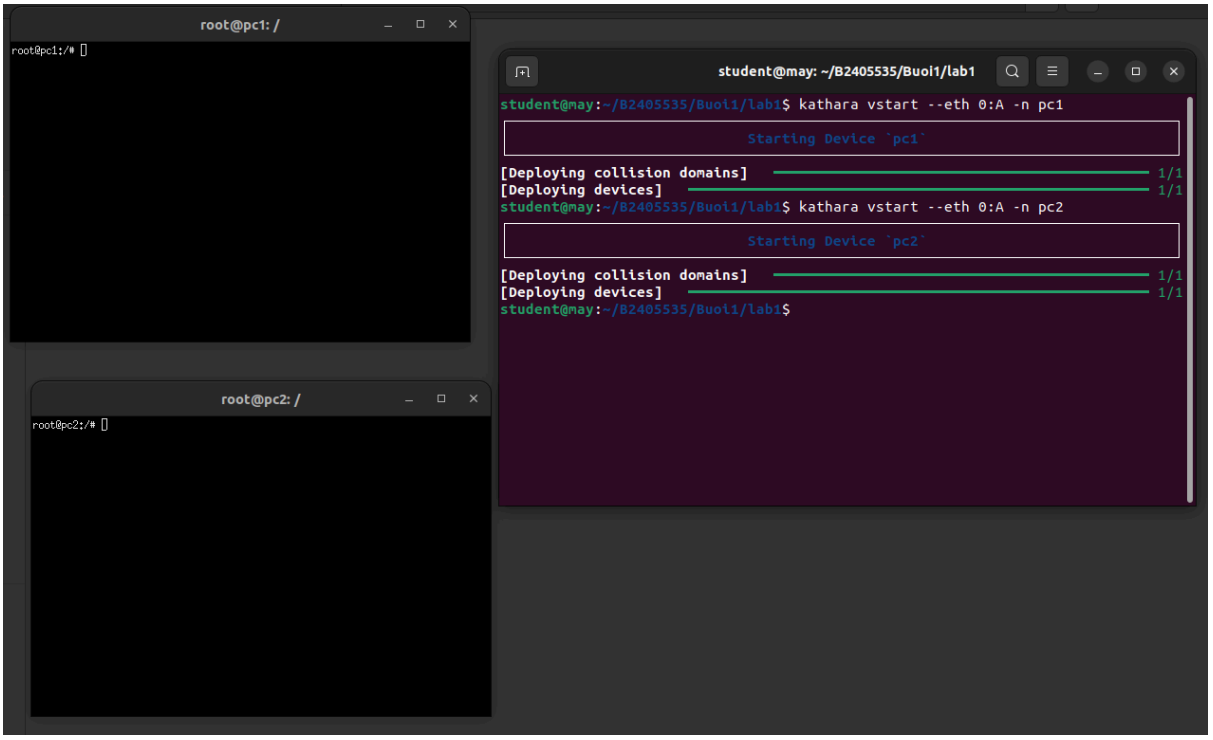
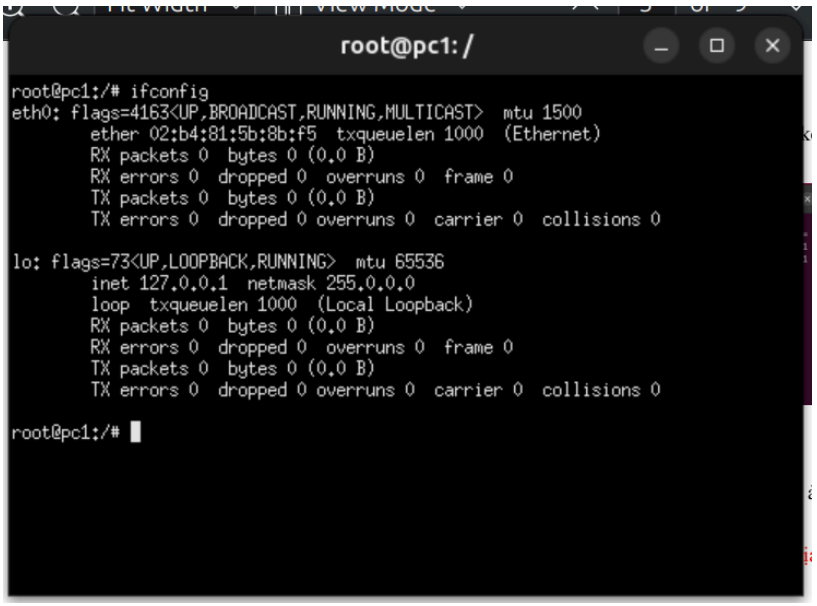
LAB 1:

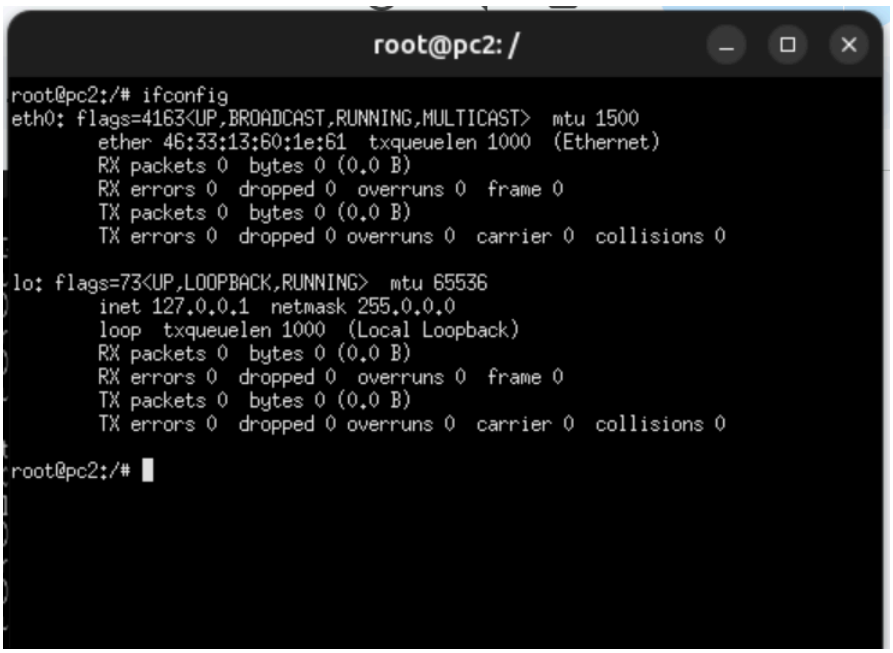
3 -> 4.



- Sau khi sử dụng lệnh kathara vstart –eth 0:A -n pc1 để khởi động pc1, kathara vstart –eth 0:A -n pc2 để khởi động pc2.

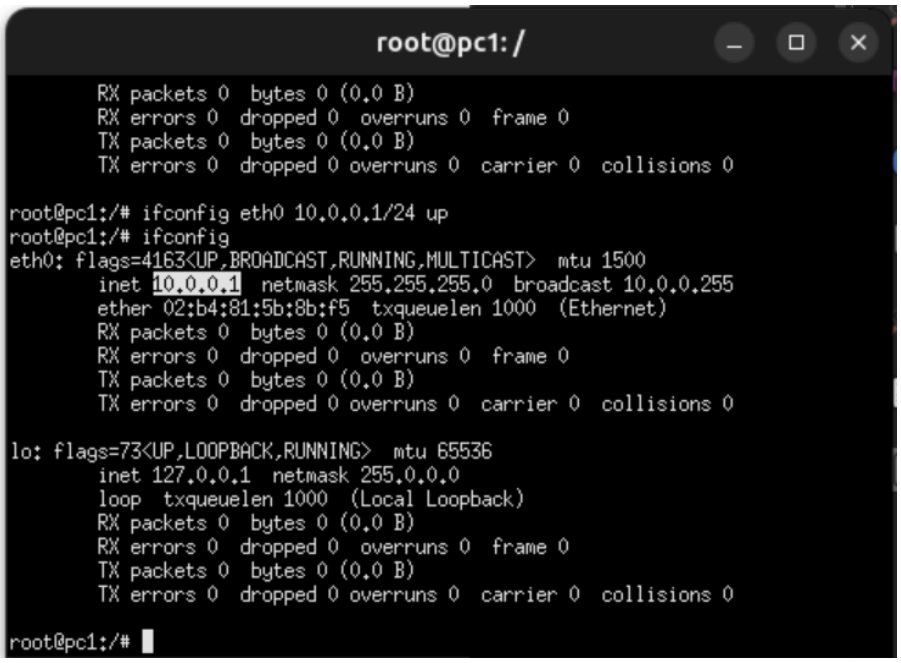
5.





- Thực hiện lênh ifconfig trên cả pc1 và pc2. Do chưa gán địa chỉ IP nên pc1 và pc2 lần lượt có các địa chỉ MAC là: 02:b4:81:5b:8b:f5 và 46:33:12:60:1e:61
- Những giao diện được tạo trong máy ảo: eth0, lo

6.



Thực hiên lệnh gán địa chỉ IP trên pc1: ifconfig eth0 10.0.0.1/24, ta được địa chỉ IP như hình trên.

7.

```
root@pc2: /
        RX packets 0  bytes 0 (0.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 0  bytes 0 (0.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

root@pc2:/# ifconfig eth0 10.0.0.2/24
root@pc2:/# ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.0.0.2  netmask 255.255.255.0  broadcast 10.0.0.255
        ether 46:33:13:60:1e:61  txqueuelen 1000  (Ethernet)
        RX packets 0  bytes 0 (0.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 0  bytes 0 (0.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 0  bytes 0 (0.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 0  bytes 0 (0.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

root@pc2:/#
```

Thực hiên lệnh gán địa chỉ IP trên pc1: ifconfig eth0 10.0.0.1/24, ta được địa chỉ IP như hình trên.

8.

```
root@pc1: /
        RX packets 0  bytes 0 (0.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 0  bytes 0 (0.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

root@pc1:/# ping 10.0.0.2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=0.508 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.329 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.258 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.287 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.276 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=0.373 ms
64 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=0.294 ms
64 bytes from 10.0.0.2: icmp_seq=8 ttl=64 time=0.275 ms
64 bytes from 10.0.0.2: icmp_seq=9 ttl=64 time=0.281 ms
64 bytes from 10.0.0.2: icmp_seq=10 ttl=64 time=0.285 ms
64 bytes from 10.0.0.2: icmp_seq=11 ttl=64 time=0.275 ms
64 bytes from 10.0.0.2: icmp_seq=12 ttl=64 time=0.511 ms
^C
--- 10.0.0.2 ping statistics ---
12 packets transmitted, 12 received, 0% packet loss, time 11267ms
rtt min/avg/max/mdev = 0.258/0.329/0.511/0.085 ms
root@pc1:/#
```

Trên pc1, thực hiện lệnh ping lên máy tính đích pc2.

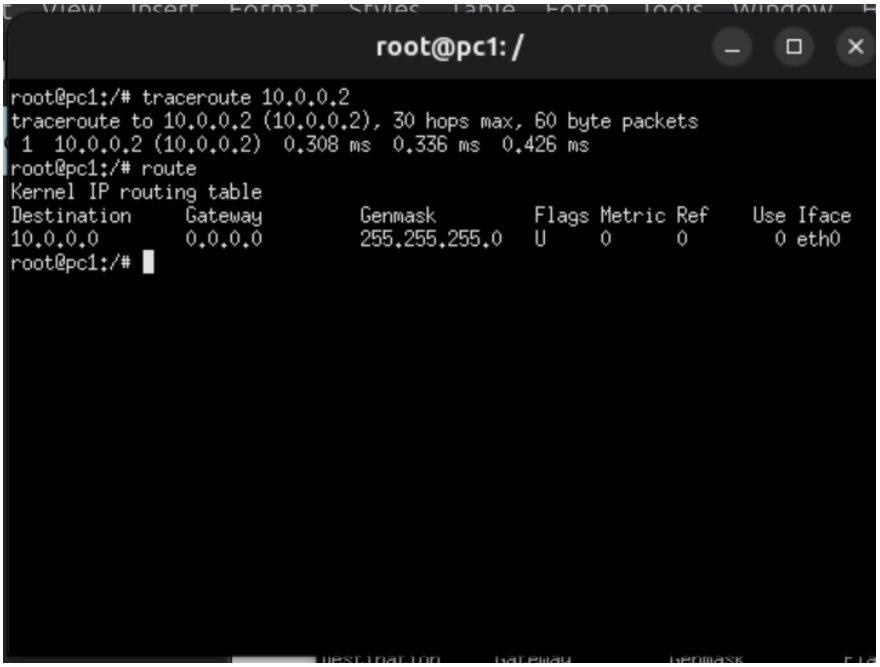
-> Ý nghĩa: Đây là lệnh gửi gói tin ICMP từ máy nguồn pc1 đến máy đích pc2, pc2 tồn tại nên gửi lại gói tin hồi đáp ngược lại pc1.

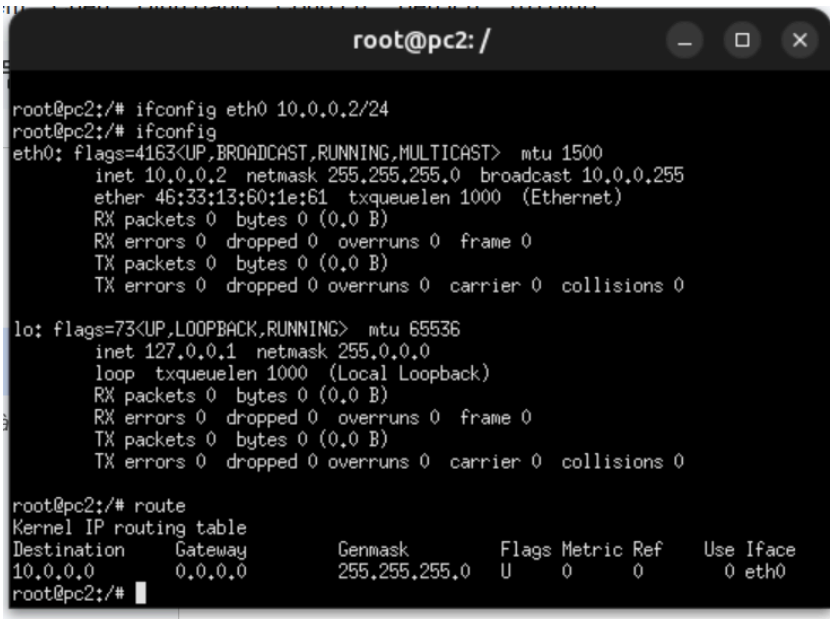
9.

- Sử dụng lệnh traceroute từ pc1 lên pc2

-> Kết quả: pc1 kết nối trực tiếp với pc2 thông qua 1 hop.

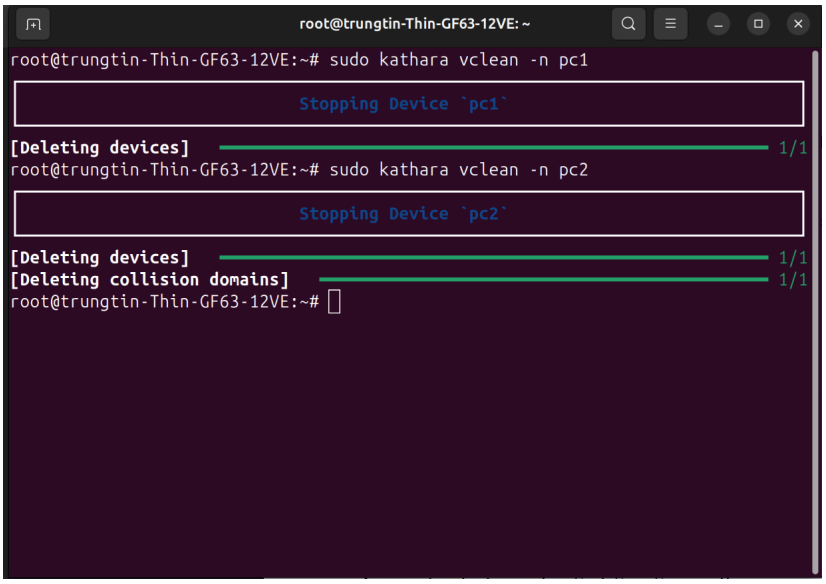
- Sử dụng lệnh route:





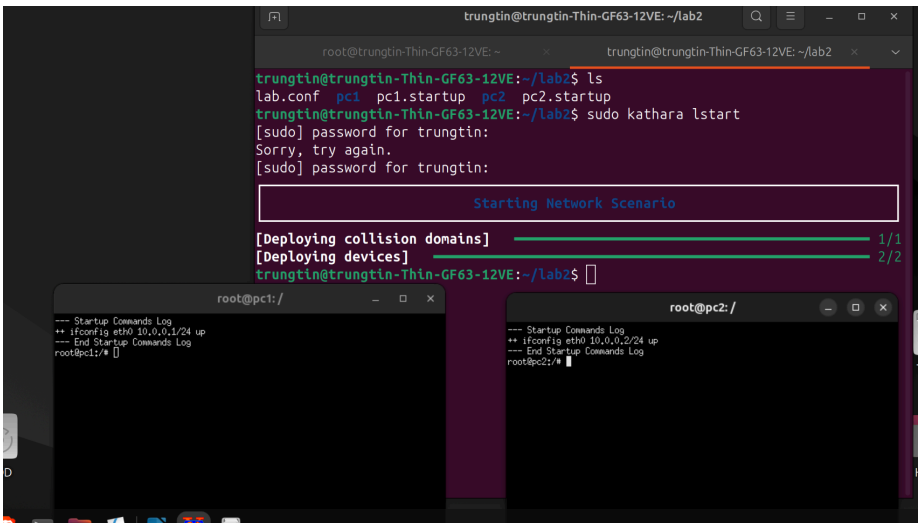
-> kết quả trên cho biết bảng vạch đường đi của pc1 và pc2, cho biết đường đi của các gói tin pc1 và pc2 trong mạng.

10.

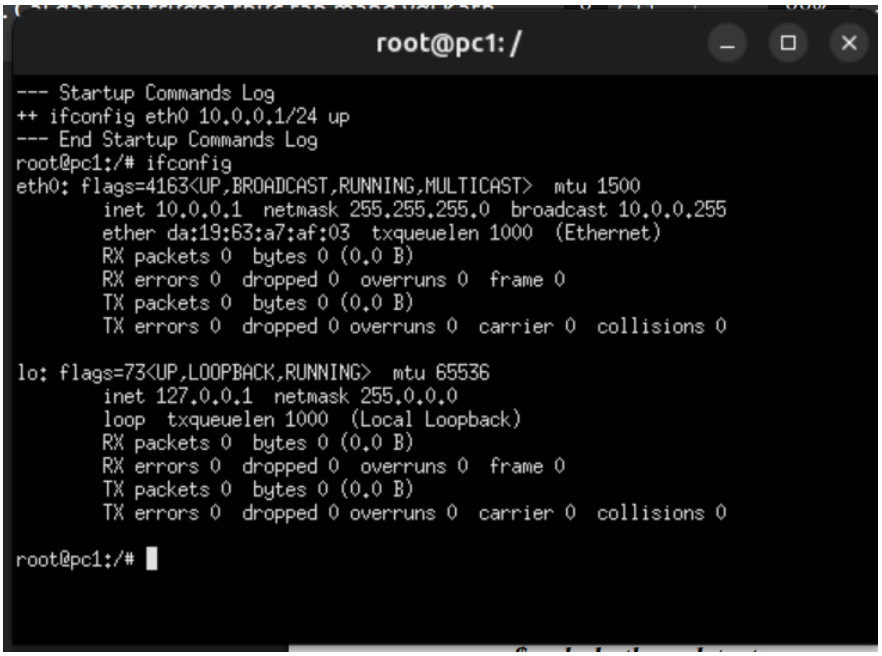


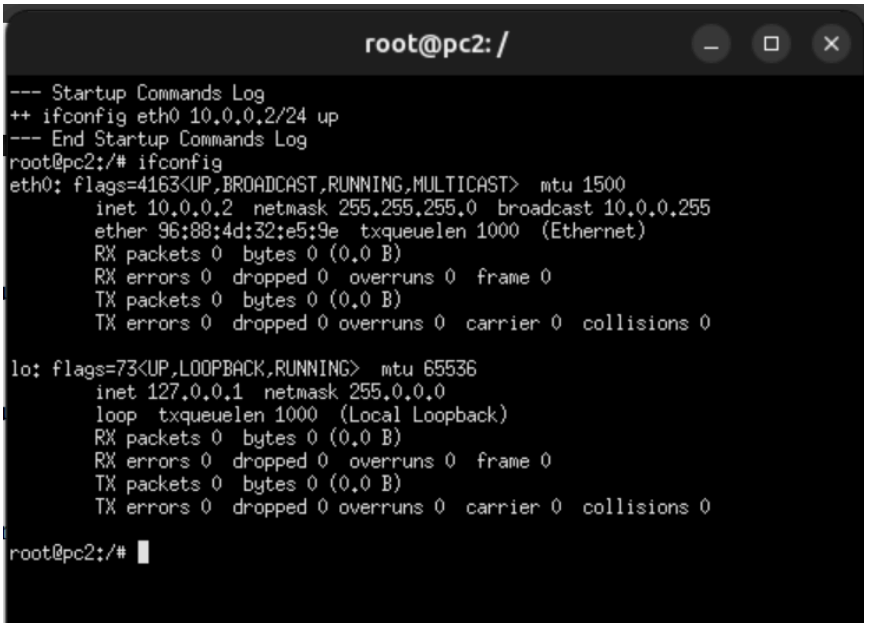
LAB 2:

1 -> 5.


trungtin@trungtin-Thin-GF63-12VE: ~/lab2
root@trungtin-Thin-GF63-12VE: ~
trungtin@trungtin-Thin-GF63-12VE: ~/lab2
trungtin@trungtin-Thin-GF63-12VE:~/lab2$ ls
lab.conf pc1 pc1.startup pc2 pc2.startup
trungtin@trungtin-Thin-GF63-12VE:~/lab2$ sudo kathara lstart
[sudo] password for trungtin:
Sorry, try again.
[sudo] password for trungtin:
Starting Network Scenario
[Deploying collision domains] 1/1
[Deploying devices] 2/2
trungtin@trungtin-Thin-GF63-12VE:~/lab2$
root@pc1: /
--- Startup Commands Log
++ ifconfig eth0 10.0.0.1/24 up
--- End Startup Commands Log
root@pc1:/#
root@pc2: /
--- Startup Commands Log
++ ifconfig eth0 10.0.0.2/24 up
--- End Startup Commands Log
root@pc2:/#

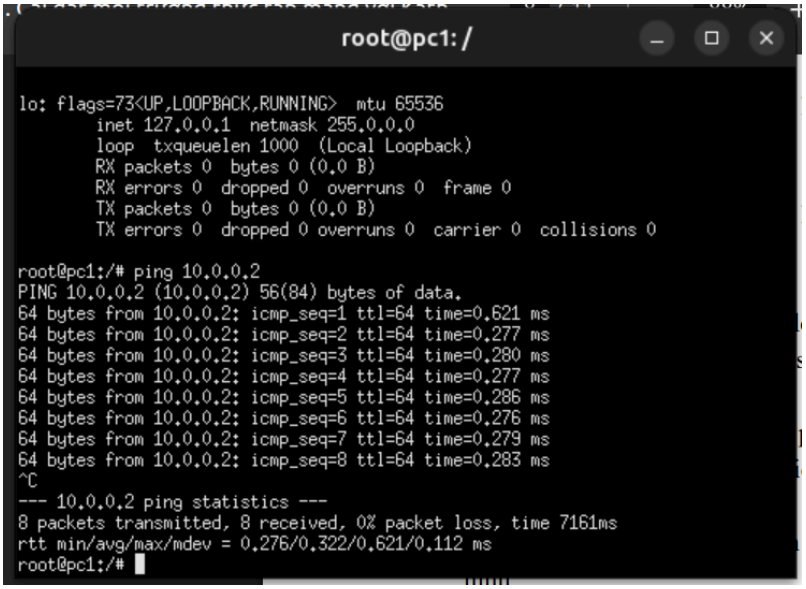
6.


root@pc1: /
--- Startup Commands Log
++ ifconfig eth0 10.0.0.1/24 up
--- End Startup Commands Log
root@pc1:/# ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
inet 10.0.0.1 netmask 255.255.255.0 broadcast 10.0.0.255
ether da:19:63:a7:af:03 txqueuelen 1000 (Ethernet)
RX packets 0 bytes 0 (0.0 B)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 0 bytes 0 (0.0 B)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
inet 127.0.0.1 netmask 255.0.0.0
loop txqueuelen 1000 (Local Loopback)
RX packets 0 bytes 0 (0.0 B)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 0 bytes 0 (0.0 B)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
root@pc1:/#


root@pc2: /
--- Startup Commands Log
++ ifconfig eth0 10.0.0.2/24 up
--- End Startup Commands Log
root@pc2:/# ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
inet 10.0.0.2 netmask 255.255.255.0 broadcast 10.0.0.255
ether 96:88:4d:32:e5:9e txqueuelen 1000 (Ethernet)
RX packets 0 bytes 0 (0.0 B)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 0 bytes 0 (0.0 B)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
inet 127.0.0.1 netmask 255.0.0.0
loop txqueuelen 1000 (Local Loopback)
RX packets 0 bytes 0 (0.0 B)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 0 bytes 0 (0.0 B)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
root@pc2:/#

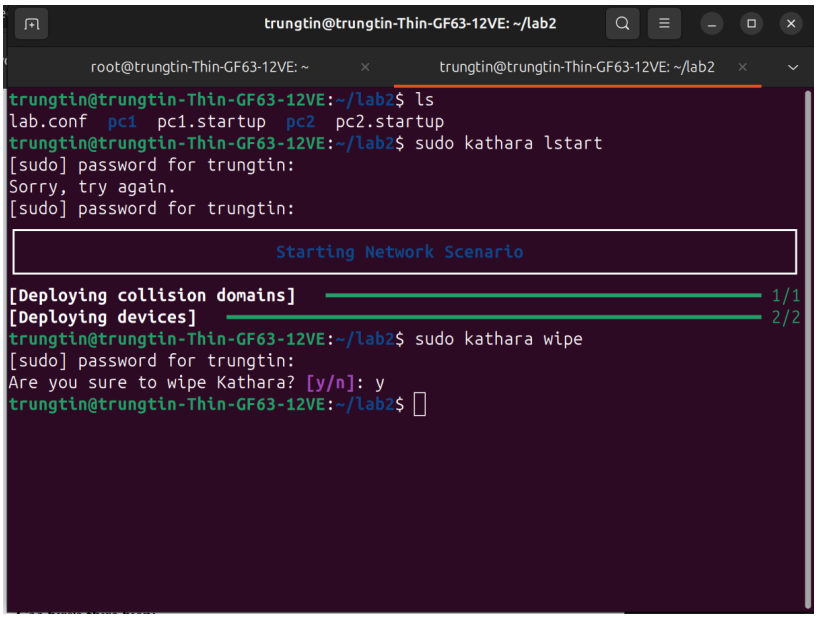
Sau khi tạo 2 máy ảo, thực hiện lệnh ifconfig để kiểm tra IP.

7.



Trên pc1, thực hiện lệnh ping đến pc2.

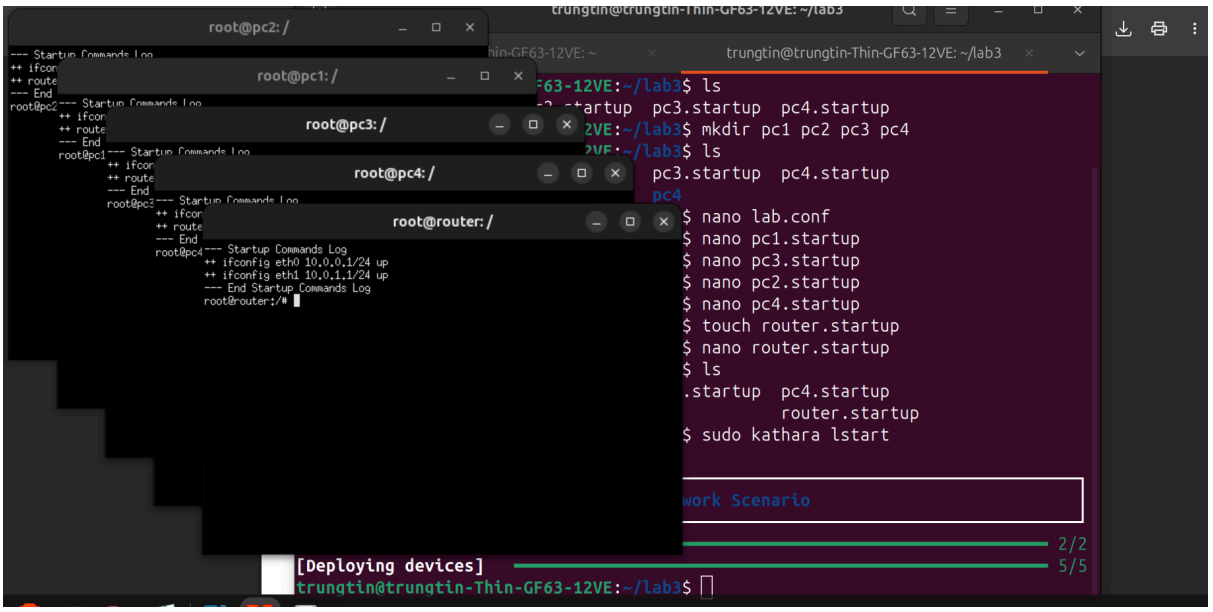
8.



Đóng 2 máy ảo lại

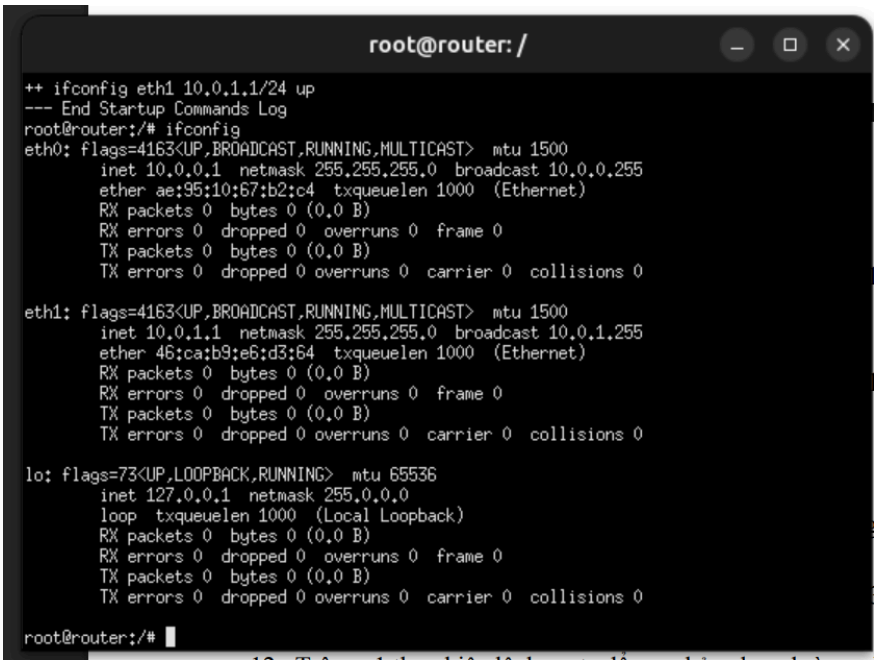
LAB 3:

9.



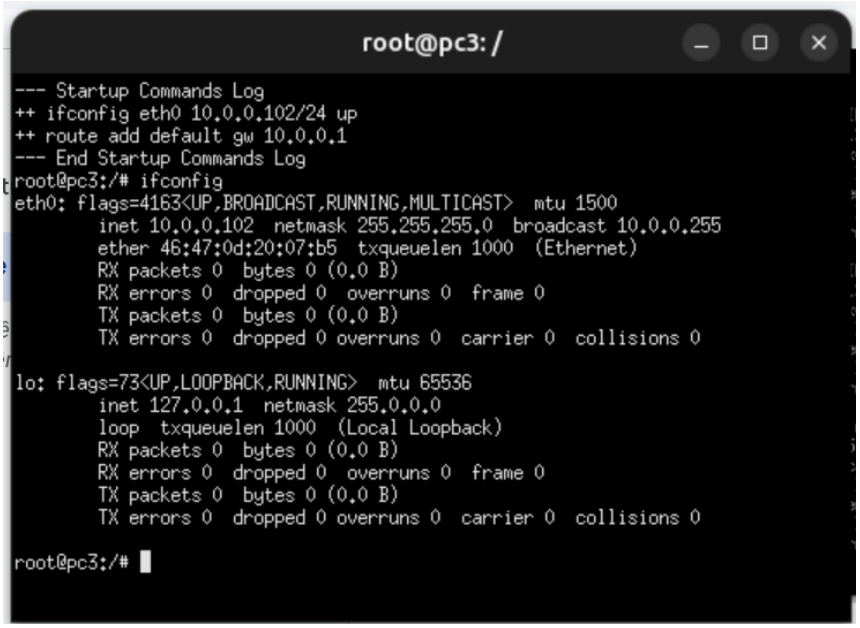
10.

Thực hiện lệnh ifconfig trên pc1, pc2, pc3, pc4 và router.

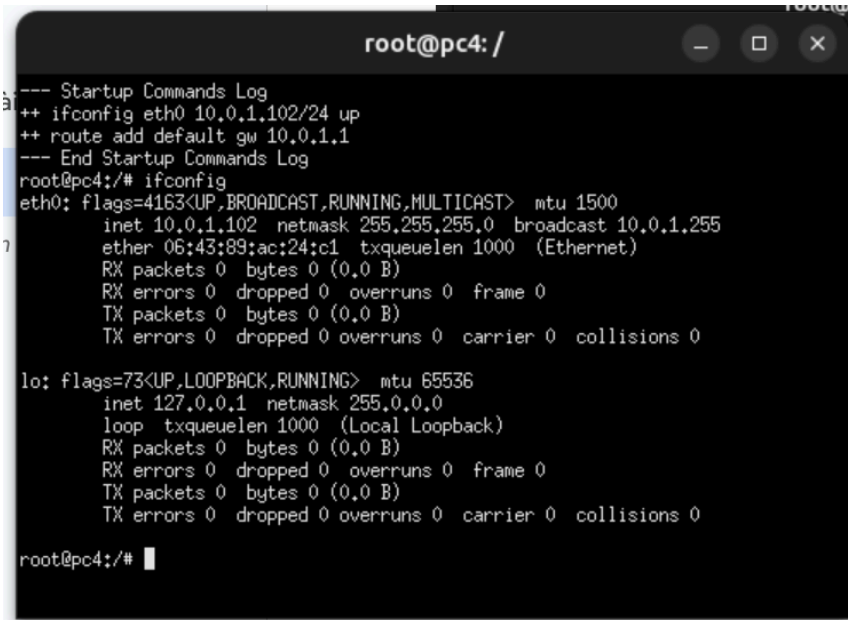


```
root@pc1: /
--- Startup Commands Log
++ ifconfig eth0 10.0.0.101/24 up
++ route add default gw 10.0.0.1
--- End Startup Commands Log
root@pc1:/# ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.0.0.101  netmask 255.255.255.0  broadcast 10.0.0.255
        ether 0e:73:d7:ba:c7:a6  txqueuelen 1000  (Ethernet)
        RX packets 0  bytes 0 (0.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 0  bytes 0 (0.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 0  bytes 0 (0.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 0  bytes 0 (0.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

root@pc1:/#
```

```
root@pc2: /
--- Startup Commands Log
++ ifconfig eth0 10.0.1.101/24 up
++ route add default gw 10.0.1.1
--- End Startup Commands Log
root@pc2:/# ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.0.1.101  netmask 255.255.255.0  broadcast 10.0.1.255
        ether b6:d2:1d:90:d6:c8  txqueuelen 1000  (Ethernet)
        RX packets 0  bytes 0 (0.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 0  bytes 0 (0.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 0  bytes 0 (0.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 0  bytes 0 (0.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

root@pc2:/#
```



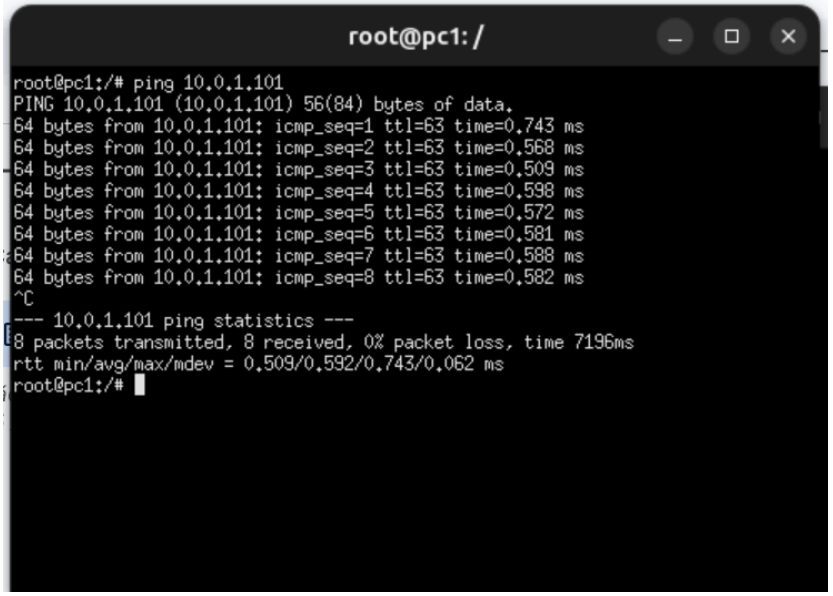
```
root@pc3: /
--- Startup Commands Log
++ ifconfig eth0 10.0.0.102/24 up
++ route add default gw 10.0.0.1
--- End Startup Commands Log
root@pc3:/# ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.0.0.102  netmask 255.255.255.0  broadcast 10.0.0.255
        ether 46:47:0d:20:07:b5  txqueuelen 1000  (Ethernet)
        RX packets 0  bytes 0 (0.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 0  bytes 0 (0.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 0  bytes 0 (0.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 0  bytes 0 (0.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

root@pc3:/#
```



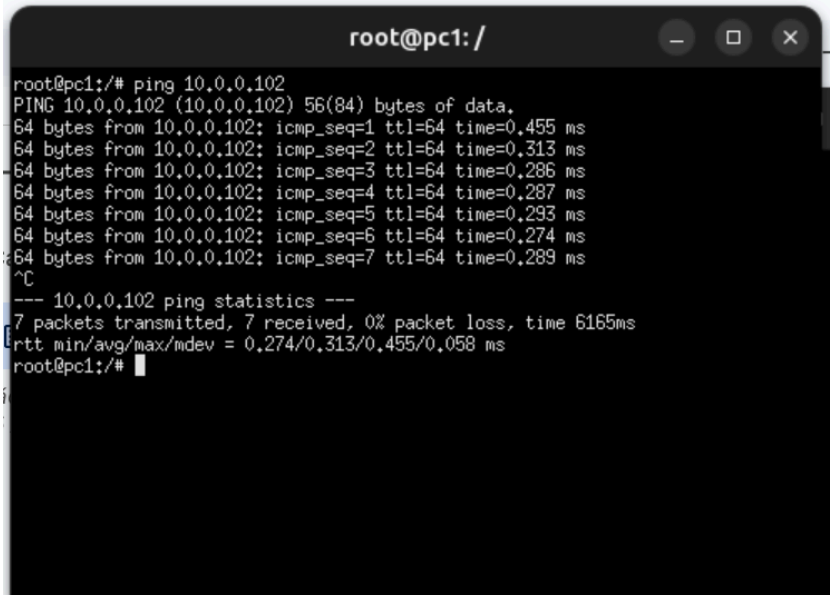
```
root@pc4: /
--- Startup Commands Log
++ ifconfig eth0 10.0.1.102/24 up
++ route add default gw 10.0.1.1
--- End Startup Commands Log
root@pc4:/# ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.0.1.102  netmask 255.255.255.0  broadcast 10.0.1.255
        ether 06:43:89:ac:24:c1  txqueuelen 1000  (Ethernet)
        RX packets 0  bytes 0 (0.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 0  bytes 0 (0.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 0  bytes 0 (0.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 0  bytes 0 (0.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

root@pc4:/#
```

11.

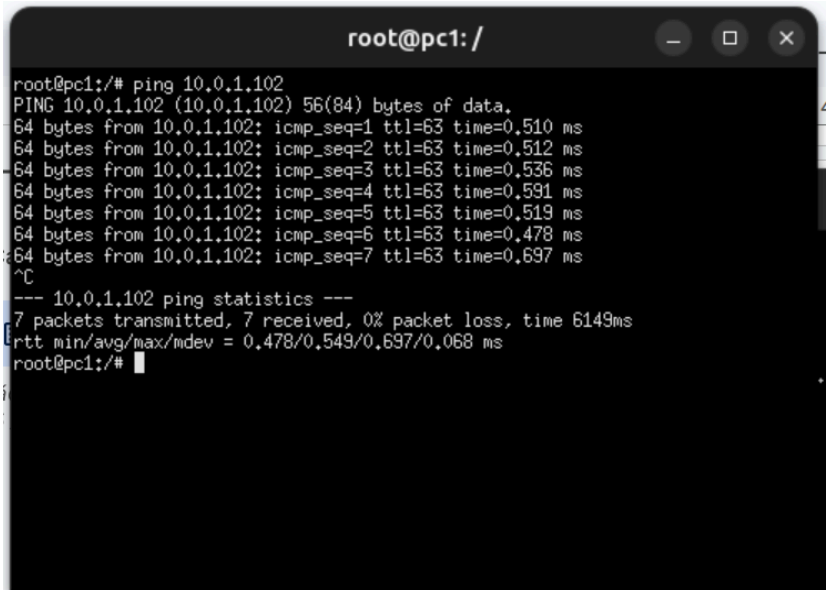
- pc1 ping pc2:



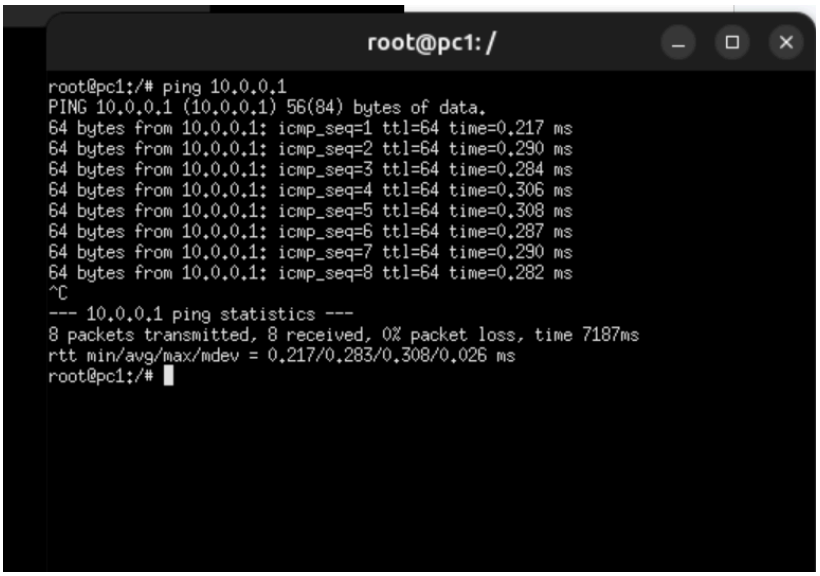
- pc1 ping pc3:



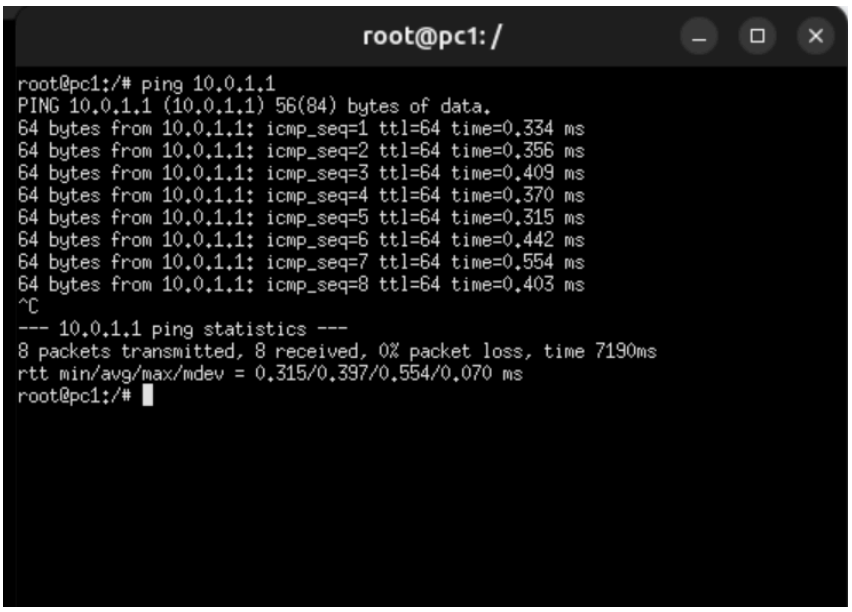
- pc1 ping pc4:



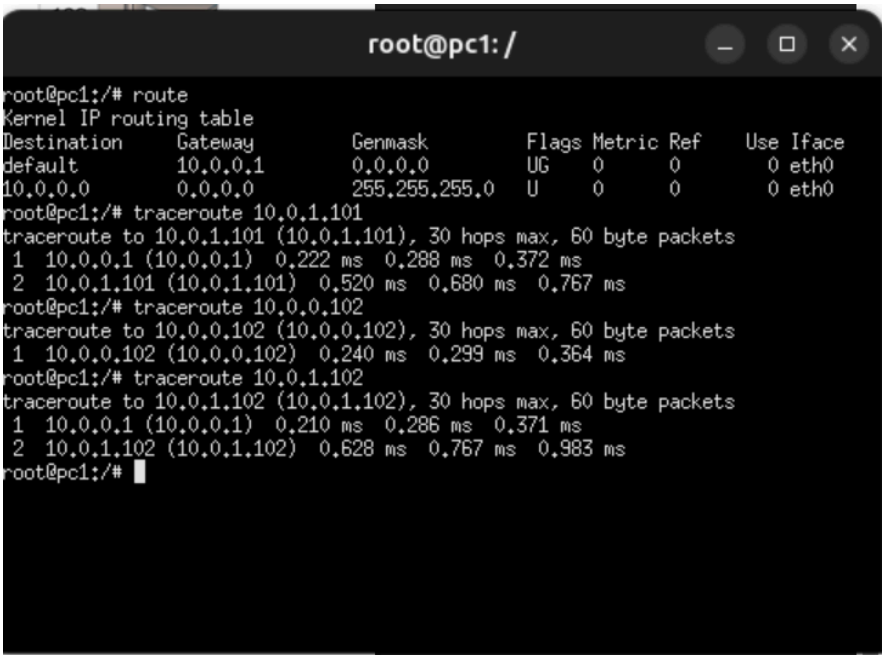
- pc1 ping eth0:



- pc1 ping eth1:



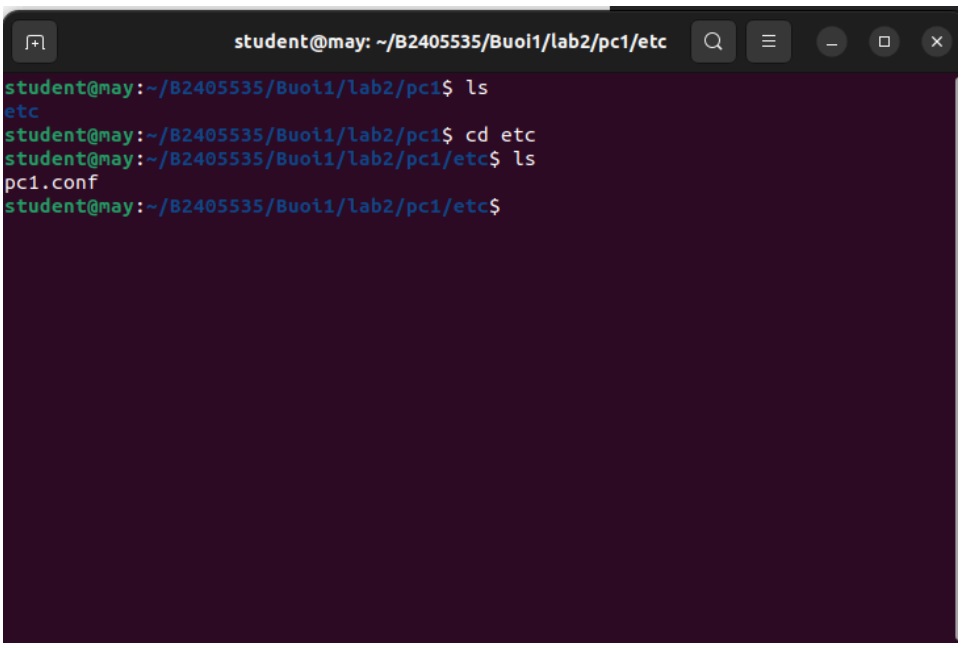
12.

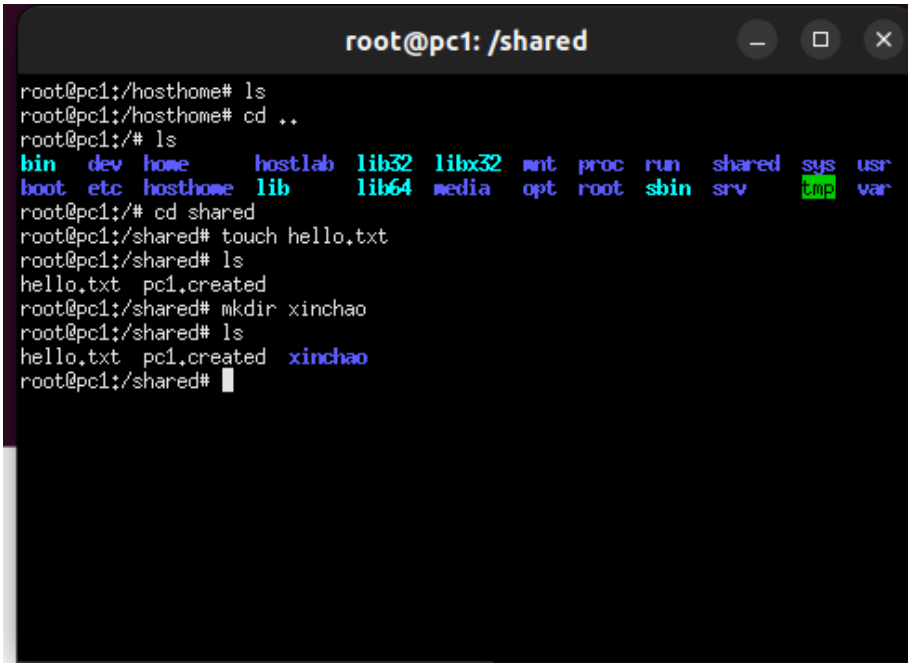


Trên pc1:

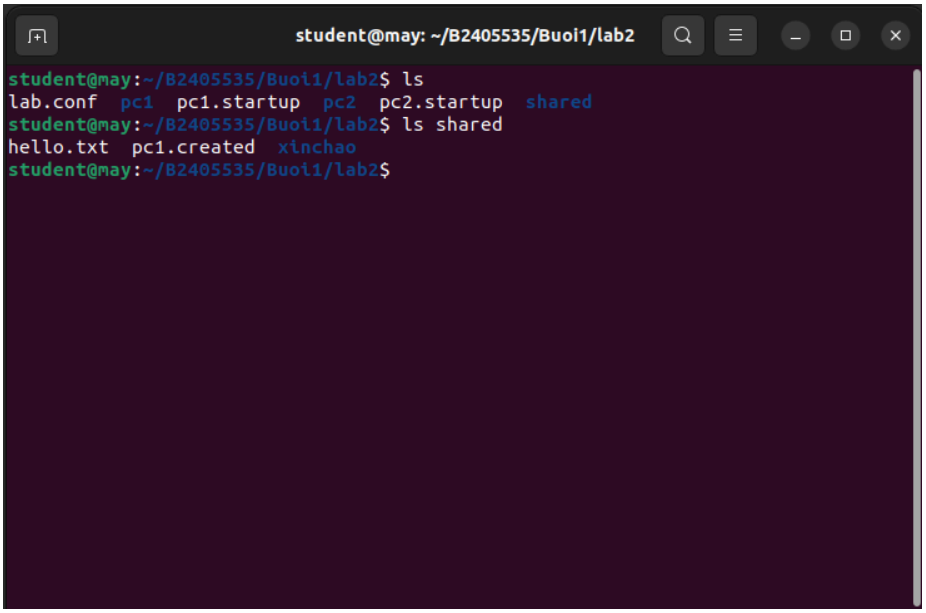
- Thực hiện lệnh route, kết quả sẽ ra:
  + default: cổng mặc định -> dùng để đi ra các địa chỉ nằm ngoài mạng cục bộ
  + 10.0.0.0: Dãy mạng cục bộ, dùng để pc1 kết nối trực tiếp các máy tính có dải địa chỉ cục bọ từ 10.0.0.1 -> 10.0.0.254
- Thực hiện lệnh traceroute trên pc1 đến đia chỉ của từng pc:
  + pc2: pc1 đi qua default gateway -> địa chỉ IP pc2 (10.0.1.101)
  + pc3: pc1 đến trực tiếp pc3 (10.0.0.102)
  + pc4: pc1 đi qua default gateway -> địa chỉ IP pc4 (10.0.1.102)

LAB 4:





- Trong thư mục shared, lần lượt tạo file hello.txt, pc1.created và thư mục xinchao



- Sau khi thực hiện trên máy ảo xong. Trên máy thực kiểm tra trong thư mục shared. Kết quả hiển thị như hình trên

-> Kết quả: chia sẻ nội dung thư mục pc1 lên máy thực thành công